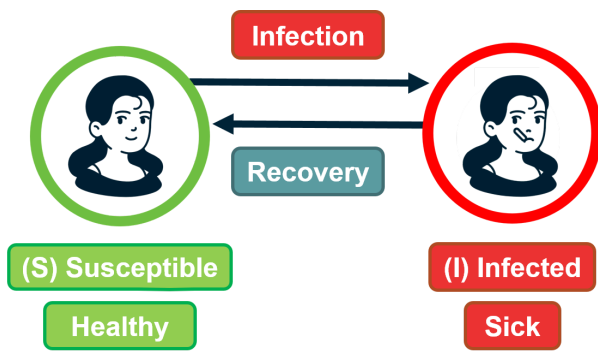


L9: SIS Model – With An Arbitrary Degree Distribution



Let us go back to the SIS model.

With the degree block approximation, we model the density of susceptible $s_k(t)$ and infected $i_k(t)$ individuals that have degree k .

Of course, it is still true that $s_k(t) + i_k(t) = 1$ because any of these individuals is either in the S or I states.

We can also write that the density of all infected individuals is:

$$i(t) = \sum_k p_k i_k(t).$$

A susceptible individual of degree k can become infected when he/she is in contact with an infected individual. For nodes of degree k , what is the fraction of their neighbors that are infected however? Under the homogeneous mixing assumption, this fraction is simply $i(t)$. We now need to derive this fraction more carefully, considering that different nodes have different degrees.

So, let us define as $\theta_k(t)$ as the fraction of infected neighbors of degree k node.

If we manage to calculate this fraction, we can then write the differential equation for the SIS model under the degree block approximation as:

$$\frac{di_k(t)}{dt} = k\beta\theta_k(t)(1 - i_k(t)) - \mu i_k(t)$$

Note that the only real difference with the SIS differential equation under homogeneous mixing is that the term $\theta_k(t)$ has replaced the term $i(t)$. The reason is that susceptible individuals of degree k -- their density is $(1 - i_k(t))$ -- get infected from a fraction $\theta_k(t)$ of their k neighbors.

Now, let us derive $\theta_k(t)$.

Suppose we have a network with n nodes, m edges, and an arbitrary degree distribution p_k .

Recall that the average degree is given by $\bar{k} = \frac{2m}{n}$, and the average number of nodes of degree k is $n_k = n p_k$.

Consider a node of degree k . The probability that a neighbor of that node has degree k' is the fraction of edge stubs in the network that belong to nodes of degree k' :

$$\frac{k' n_{k'}}{2m} = \frac{k' p_{k'}}{\bar{k}}$$

Note that this probability does not depend on k .

So, the probability that a node of degree k connects to an infected neighbor of degree k' is:

$$\frac{k' p_{k'}}{\bar{k}} i_{k'}(t)$$

Taking the summation of these probabilities across all possible values of k' we get that the probability $\theta_k(t)$ that a node of degree k connects to an infected neighbor (of any degree) is:

$$\theta_k(t) = \sum_{k'} \frac{k' p_{k'}}{\bar{k}} i_{k'}(t)$$

Note that $\theta_k(t)$ does not depend on k , and so we can simplify our notation and write $\theta(t)$ instead of $\theta_k(t)$.

This is important: the probability that any of your neighbors is infected does not depend on how many neighbors you have.

We can now go back to the original differential equation for the SIS model and re-write it as:

$$\frac{di_k(t)}{dt} = k\beta\theta(t)(1 - i_k(t)) - \mu i_k(t)$$

Early in the outbreak, when $i_k(t) \approx 0$, this (nonlinear) differentiation equation can be simplified as:

$$\frac{di_k(t)}{dt} = k\beta\theta(t) - \mu i_k(t)$$

Additionally, we should consider that early in the outbreak, an infected individual x must have one infected neighbor y (the node that infected x) -- y has not returned back to the pool of Susceptible individuals yet because we assume that we are early in the outbreak. With this correction in mind, we should modify the previous equation for the probability that a susceptible individual of degree- k gets infected from an individual of degree- k' as follows:

$$\theta_k(t) = \sum_{k'} \frac{(k'-1)p_{k'}}{\bar{k}} i_{k'}(t)$$

because one of the k' links of the infected individual must connect to another infected individual.

To solve this equation, we can take the derivative of $\theta(t)$:

$$\frac{d\theta(t)}{dt} = \sum_{k'} \frac{(k'-1)p_{k'}}{\bar{k}} \frac{di_{k'}(t)}{dt}$$

If we replace k' with k (just a notational simplification) – and substitute the derivative of $i_k(t)$ from the SIS differential equation, we get:

$$\begin{aligned} \frac{d\theta(t)}{dt} &= \sum_k \frac{(k-1)p_k}{\bar{k}} (k\beta\theta(t) - \mu i_k(t)) = \\ &= \frac{\beta}{\bar{k}} \sum_k k^2 p_k \theta(t) - \frac{\beta}{\bar{k}} \sum_k k p_k \theta(t) - \frac{\mu}{\bar{k}} \sum_k (k-1) p_k i_k(t) = \\ &= \left(\frac{\beta \bar{k}^2}{\bar{k}} - (\beta + \mu) \right) \theta(t) \end{aligned}$$

where $\bar{k}^2 = \sum_k k^2 p_k$ is the second moment of the degree distribution.

This is a linear differential equation with solution:

$$\theta(t) = c e^{t(\beta \bar{k}^2 - (\beta + \mu)\bar{k})/\bar{k}}$$

where c is a constant that depends on the initial condition.

Now that we have solved for $\theta(t)$, we could go back and derive the fraction $i_k(t)$ of infected individuals of degree k .

For our purposes, however, we do not even need to take that extra step. The expression for $\theta(t)$ clearly shows that we will have an outbreak if and only if $\beta \bar{k}^2 - (\beta + \mu) \bar{k} > 0$, or equivalently, $\frac{\beta}{\beta + \mu} > \frac{\bar{k}}{\bar{k}^2}$.

Contrast this inequality with the corresponding condition under homogeneous mixing, namely: $\beta \bar{k} - \mu > 0$, or equivalently, $\frac{\beta}{\mu} > \frac{1}{\bar{k}}$.

In other words, when we consider an arbitrary degree distribution, it is not just the average degree that affects the epidemic threshold. The second moment of the degree distribution also matters. And as the second moment increases relative to the first (i.e., the ratio $\bar{k}/\bar{k}^2$ decreases), it is easier to get an epidemic outbreak.

Food For Thought

Use the derived expression for $\theta(t)$ to derive the density $i_k(t)$ of infected individuals of degree k .